THE UNIVERSITY OF DANANG – UNIVERSITY OF SCIENCE AND TECHNOLOGY

FACULTY OF ADVANCED SCIENCE AND TECHNOLOGY

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REPORT

Title: AI-Integrated Early Fire Warning System

Supervisor: **Pham Van Tuan**

Student Name: **Nguyen Thi Van Anh**

Nguyen Dang Khoa

Pham Ho Viet Quang

Thai Trung Sinh

Class: **23ECE - Team 4**

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DESIGN THINKING

I. Definition of Design Thinking

Design Thinking is a problem-solving methodology that combines empathy for the needs of users, creativity in the generation of ideas, and rationality in the evaluation and selection of solutions. It is a human-centered approach that seeks to understand the people, their needs, and the context in which they operate, to inform the design of innovative and effective solutions.

II. Purposes of Design Thinking

Design Thinking helps us understand users more clearly, delve deeper into their problems, and better identify the challenges we must address in order to provide appropriate solutions and help our projects achieve their desired objectives.

III. Phases of Design Thinking

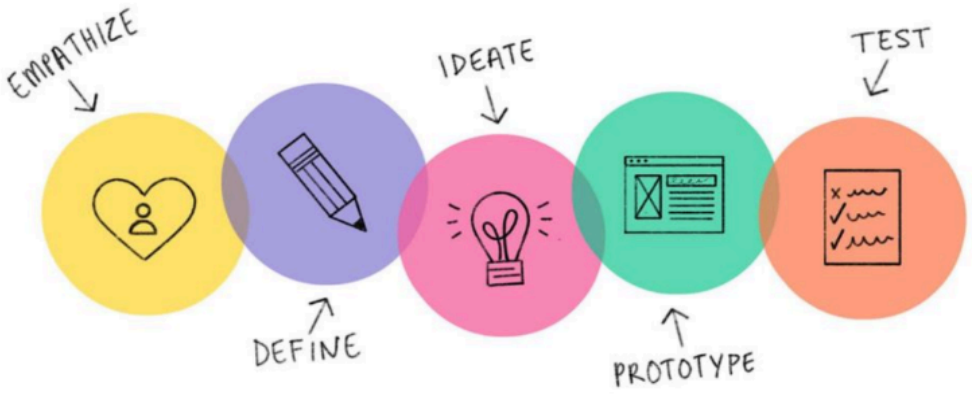


Figure 1.1: Design Thinking

Stage 1: Empathize – Research Your Users' Needs
- Gaining an empathetic awareness of the problems of people around us

Stage 2: Define – State Your Users' Needs and Problems
- Analyze our views and summarize them to determine the main issues

Stage 3: Ideate – Challenge Assumptions and Create Ideas
- Seek for different ideas on the problem and solution to it

Stage 4: Prototype – Start to Create Solutions
- Make some low-cost, scaled-down models of the product to test the ideas we've generated.

Stage 5: Test – Try Your Solutions Out
- Use outcome to establish one or more issues, return to previous stages to conduct more trials.

A. Nội Dung

Stage 1 - Thấu hiểu người dùng

1. Mục đích:

- Hiểu hệ thống phục vụ ai và cách nó được sử dụng trong môi trường thực tế
- Xác định hành vi, điểm đau và giới hạn của người dùng
- Nắm bắt các ràng buộc về môi trường và vận hành
- Đảm bảo hệ thống được thiết kế dựa trên nhu cầu thực tế, không phải giả định.

2. Các bên liên quan (Stakeholders):

- Người dùng chính: Bảo vệ – theo dõi hệ thống, nhận cảnh báo, ra quyết định trong tình huống khẩn cấp
- Người dùng phụ: Quản lý tòa nhà, nhân viên bảo trì kỹ thuật – theo dõi hiệu suất hệ thống, bảo trì và xử lý sự cố phần cứng
- Người dùng bị ảnh hưởng: Người dùng bãi xe, khách tham quan – bị ảnh hưởng bởi độ chính xác của hệ thống (báo động giả hoặc bỏ sót)
- Bên quản lý: Cơ quan an toàn phòng cháy chữa cháy – quan tâm đến độ tin cậy và tuân thủ tiêu chuẩn an toàn

3. Hồ sơ người dùng:

Hồ Sơ	**Bảo vệ** Vai trò: Quản lý bãi tầng hầm giữ xe (đặc biệt là vào ban đêm)	**Nhân viên bảo trì** Vai trò: Bảo trì và kiểm tra các thành phần hệ thống	**Người dùng bãi xe** Không tương tác trực tiếp vào hệ thống
Mục tiêu	Phát hiện sự cố nhanh, phản ứng ngay	Bảo trì, kiểm tra thiết bị	
Điểm yếu	Không thể tập trung liên tục vào tất cả camera; dễ bỏ lỡ tín hiệu cháy sớm; quá nhiều báo động giả làm mất tin tưởng	Cảm biến bị bụi và độ ẩm ảnh hưởng; khó xác định cảm biến hỏng	
Nhu cầu	Cảnh báo rõ ràng và tức thì; ít báo động giả; xác định chính xác vị trí sự cố	Chẩn đoán hệ thống dễ dàng; truy cập nhật ký sự kiện; trạng thái hệ thống rõ ràng	An toàn cá nhân; tránh hoảng loạn do báo động giả
Môi trường	Ánh sáng yếu; nhiều màn hình; ca dài		

4. **Bối cảnh sử dụng:**

- Hệ thống vận hành 24/7
- Nhân sự hạn chế vào ca đêm
- Một nhân viên có thể giám sát nhiều khu vực
- Môi trường tối, bụi, khí thải xe (CO, CO_2), ồn ào

5. **Vấn đề người dùng:**

- Theo dõi thủ công có thể phát hiện cháy quá muộn
- Thiếu cảnh báo sớm làm tăng thiệt hại
- Không có hệ thống cảnh báo và ghi nhật ký tập trung
- Khó xác định nhanh vùng nguy hiểm
- Lo ngại về quyền riêng tư camera

6. **Bằng chứng về nhu cầu**

Trong quá trình quan sát môi trường tầng hầm mô phỏng, các điều kiện được ghi nhận:

- Mức độ ánh sáng đo được dưới 50 lux ở hầu hết các khu vực trong điều kiện ban đêm.
- Nồng độ khí thải xe (CO) có thể phát hiện được trong vòng 2 phút kể từ khi động cơ hoạt động trong không gian kín.
- Một bảo vệ được quan sát đang quản lý nhiều màn hình theo dõi cùng lúc, làm giảm sự chú ý vào từng khu vực.

→ **Hỗ trợ nhu cầu:** Phát hiện tự động để bù đắp cho sự chú ý hạn chế của con người; hệ thống phải chịu được khí thải và hoạt động trong điều kiện ánh sáng yếu.

7. **Nhu cầu người dùng:**

- Phát hiện cháy sớm; cảnh báo nhanh và rõ ràng
- Hệ thống giám sát liên tục 24/7
- Giao diện đơn giản
- Ghi nhật ký sự kiện tự động

8. **Ràng buộc bảo trì:**

- Cảm biến bị bụi bám
- Camera có thể bị che khuất
- Cần hiệu chỉnh định kỳ
- Nhân viên phải dễ dàng truy cập chẩn đoán hệ thống

9. **Ràng buộc vận hành thực tế:**

- Khí thải xe và bụi có thể gây cảnh báo giả
- Độ chính xác cảm biến phụ thuộc vào vị trí đặt và bảo trì
- Mất mạng có thể làm trễ cảnh báo từ xa
- Mất điện có thể vô hiệu hóa hệ thống
- Không thể test bằng lửa thật, phải dùng mô phỏng an toàn

Stage 2 - Định nghĩa

1. Vấn đề cần giải quyết:

Thiết kế hệ thống phát hiện và cảnh báo cháy tự động quy mô nhỏ để mô phỏng điều kiện nguy cơ cháy trong bãi đỗ xe ngầm .

Mục tiêu của hệ thống:

- Phát hiện các dấu hiệu cháy sớm như khói, khí gas, tín hiệu lửa và sự thay đổi nhiệt độ đơn giản.
- Cung cấp cảnh báo rõ ràng và nhanh chóng cho nhân viên an ninh và ban quản lý tòa nhà.
- Hoạt động liên tục (24/7) để giám sát thời gian thực.
- Giảm báo động giả do các yếu tố môi trường như bụi, độ ẩm hoặc khí thải xe cộ.
- Cho phép nhân viên an ninh nhanh chóng xác định vị trí có nguy cơ cháy.

2. Các chỉ số đánh giá thành công:

- Cảnh báo trong vòng 20 giây sau khi phát hiện tín hiệu bất thường
- Độ chính xác phát hiện ≥ 75–80%
- Tỷ lệ báo động giả ≤ 5%
- Uptime ≥ 75–80%
- Toàn bộ sự kiện được ghi nhật ký có timestamp

3. Phạm vi hệ thống:

- Yêu cầu chức năng (Hệ thống thực hiện những gì):
  + Giám sát điều kiện môi trường trong tầng hầm đỗ xe bằng cảm biến
  + Phát hiện sớm các dấu hiệu nguy cơ cháy (khói, khí gas, thay đổi nhiệt độ, tín hiệu ngọn lửa)
  + Xử lý dữ liệu cảm biến để xác định các điều kiện bất thường
  + Tạo cảnh báo tự động (chuông báo, đèn báo hoặc thông báo trên giao diện)
  + Cung cấp thông tin về vị trí của nguy cơ được phát hiện
  + Ghi lại các sự kiện, cảnh báo của hệ thống vào cơ sở dữ liệu để giám sát và phân tích
- Các yêu cầu phi chức năng (Cách hệ thống hoạt động):
  + Độ trễ phát hiện ≤ 20 giây sau khi phát hiện tín hiệu bất thường
  + Độ chính xác phát hiện ≥ 75–80% trong điều kiện mô phỏng
  + Tỷ lệ báo động sai ≤ 5% khi có nhiễu loạn môi trường (bụi, khí thải)
  + Thời gian hoạt động của hệ thống ≥ 75–80% trong quá trình thử nghiệm
  + Hệ thống phải hoạt động liên tục (khả năng giám sát 24/7)
  + Hệ thống phải hoạt động trong môi trường thiếu sáng
  + Hệ thống phải có khả năng chống chịu bụi, độ ẩm và khí thải xe cộ
  + Dữ liệu được lưu trữ trong cơ sở dữ liệu phải nhất quán, đồng bộ hóa thời gian và đáng tin cậy
- Giới hạn hệ thống + Ràng buộc hệ thống:
  + Hệ thống không tự dập lửa
  + Không thay thế hệ thống chống cháy chính thức

+ Không liên hệ trực tiếp cảnh sát PCCC
+ Chỉ là công cụ cảnh báo sớm hỗ trợ

4. Thông số Kỹ thuật Hệ thống:

- Hệ thống được triển khai dưới dạng mô hình nguyên mẫu quy mô nhỏ để mô phỏng phát hiện cháy trong môi trường bãi đỗ xe ngầm, bao gồm các thành phần và yêu cầu sau:

a. Cảm biến:

Hệ thống sử dụng 4 loại cảm biến: khói (phát hiện hạt bụi/khói), khí gas (nồng độ CO và các khí tương tự), ngọn lửa (tín hiệu lửa trực tiếp) và nhiệt độ (biến đổi nhiệt bất thường)

b. Điều kiện vận hành:

Hệ thống phải hoạt động được trong môi trường tối hoặc ánh sáng yếu, có bụi và khói, không gian hạn chế, đồng thời giám sát liên tục theo thời gian thực.

c. Bộ xử lý:

Vi điều khiển hoặc bộ nhúng thu thập dữ liệu cảm biến, phân tích điều kiện bất thường và quyết định có kích hoạt cảnh báo hay không.

d. Hệ thống cảnh báo:

Khi phát hiện điều kiện nguy hiểm, hệ thống kích hoạt còi, đèn cảnh báo hoặc thông báo trên giao diện để thông báo ngay cho nhân viên bảo vệ.

e. Giao diện giám sát:

Cho phép nhân viên theo dõi trạng thái hệ thống và nhận diện cảnh báo một cách nhanh chóng, dễ dàng.

f. Ghi nhật ký & Cơ sở dữ liệu:

Toàn bộ sự kiện cảnh báo được ghi tự động vào cơ sở dữ liệu có cấu trúc, lưu trữ các thông tin: ID cảm biến, loại cảm biến, timestamp, giá trị đo, trạng thái hệ thống, thông tin cảnh báo và nhãn dữ liệu (Bình thường / Khói / Cháy). Dữ liệu được thu thập trong 3 tình huống (bình thường, có khói, có lửa) và 3 điều kiện ánh sáng (sáng, tối, tối có đèn vàng).

g. Năng lực hệ thống:

Phát hiện các mẫu bất thường liên quan đến nguy cơ cháy; cảnh báo nhanh hơn giám sát thủ công; giảm báo động giả do nhiễu môi trường; hỗ trợ giám sát và ra quyết định theo thời gian thực.

h. Ràng buộc hệ thống:

Môi trường tối trong mô hình kín; bụi và khói ảnh hưởng đến cảm biến; không gian hạn chế trong mô hình chữ nhật quy mô nhỏ; giới hạn về độ chính xác và hiệu chỉnh cảm biến.

Phân Giải Chức Năng

Bước 1: Xác định ranh giới hệ thống bằng Mô hình Hộp đen

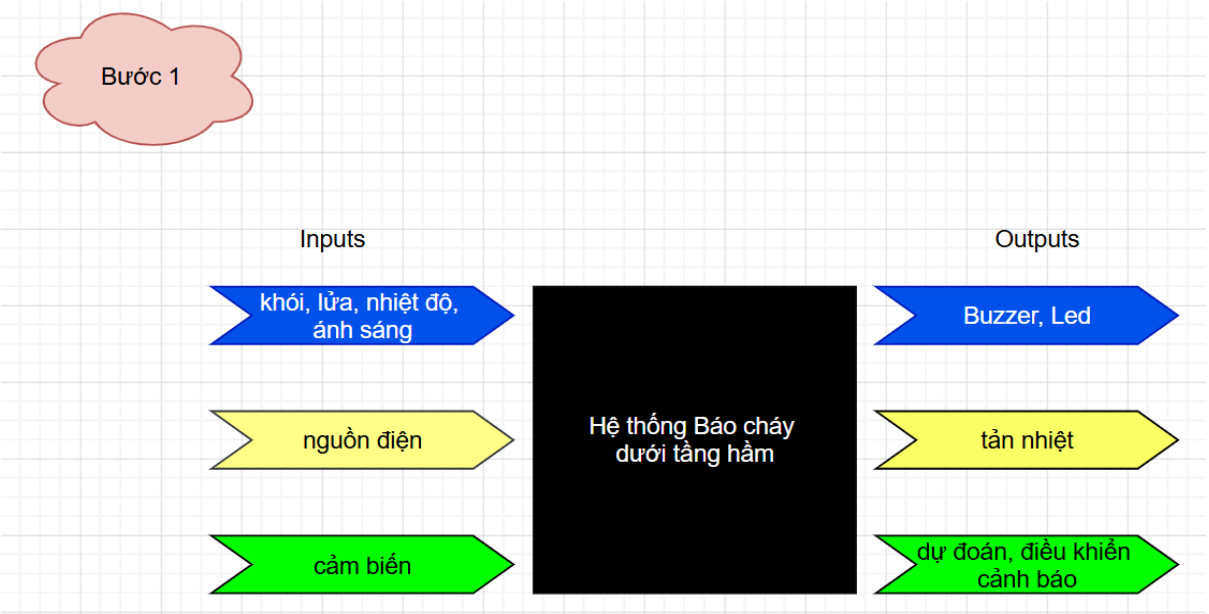


Figure 1.2: Xác định ranh giới hệ thống bằng Mô hình Hộp đen

Bước 2: Chia nhỏ các mục tiêu cốt lõi thành cây phân rã chức năng.

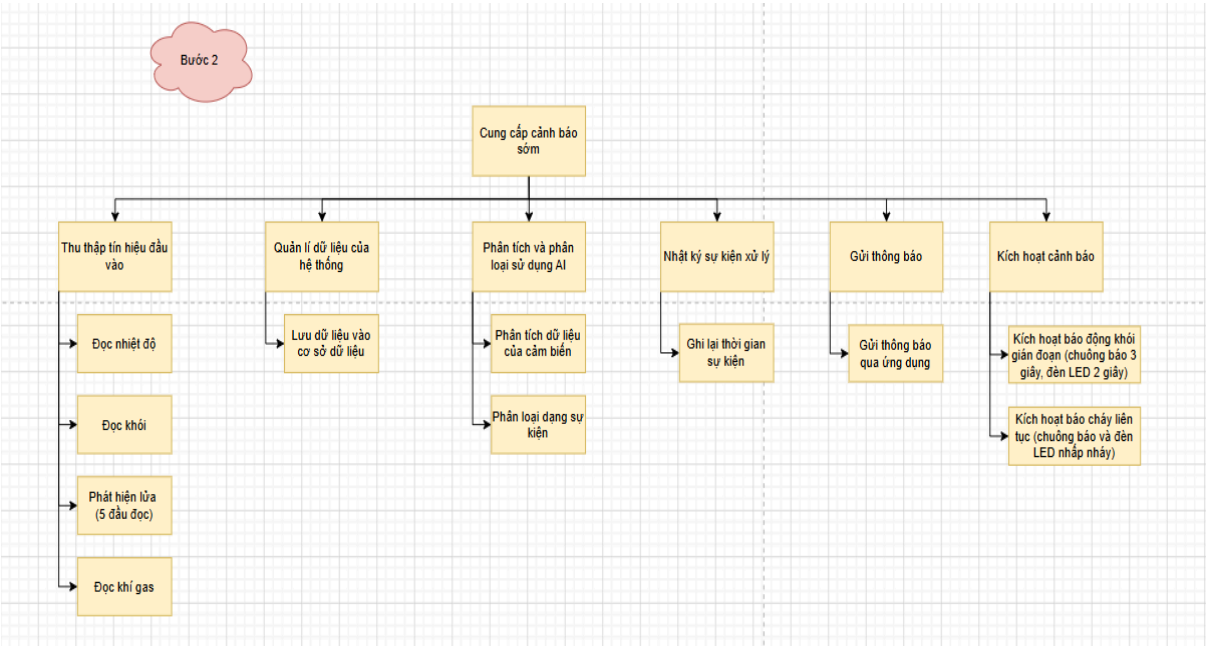


Figure 1.3: Chia nhỏ các mục tiêu cốt lõi thành cây phân rã chức năng.

Bước 3: Định tuyến chức năng bằng cách sử dụng Sơ đồ cấu trúc chức năng

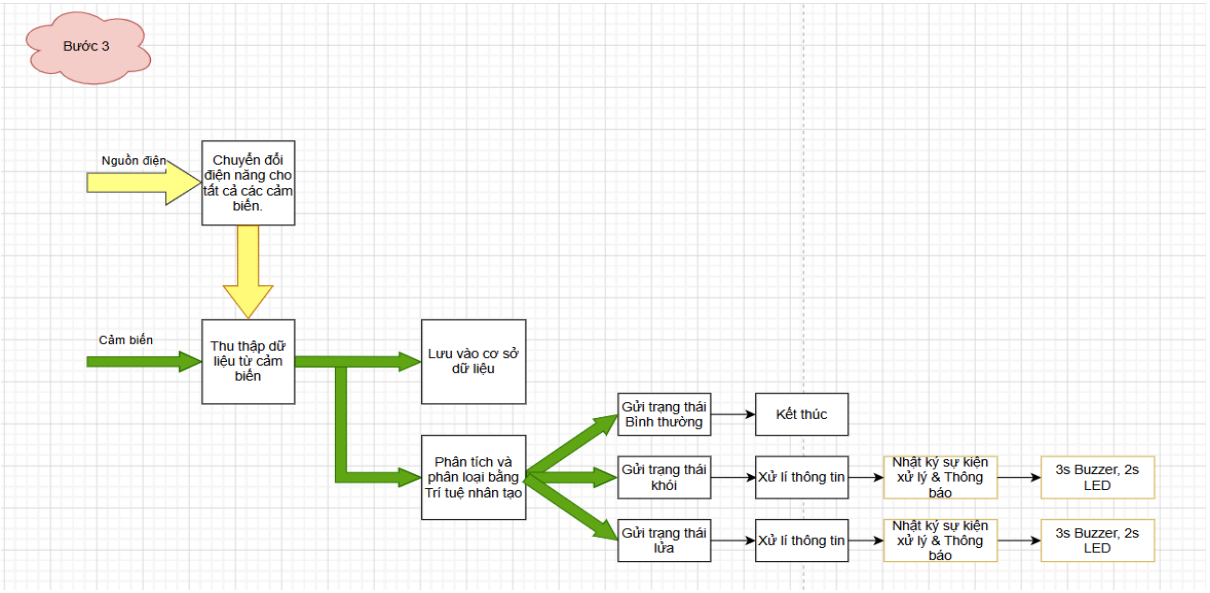


Figure 1.4: Định tuyến chức năng bằng cách sử dụng Sơ đồ cấu trúc chức năng

Bước 4: Ánh xạ các chức năng riêng lẻ tới các giải pháp vật lý khả thi

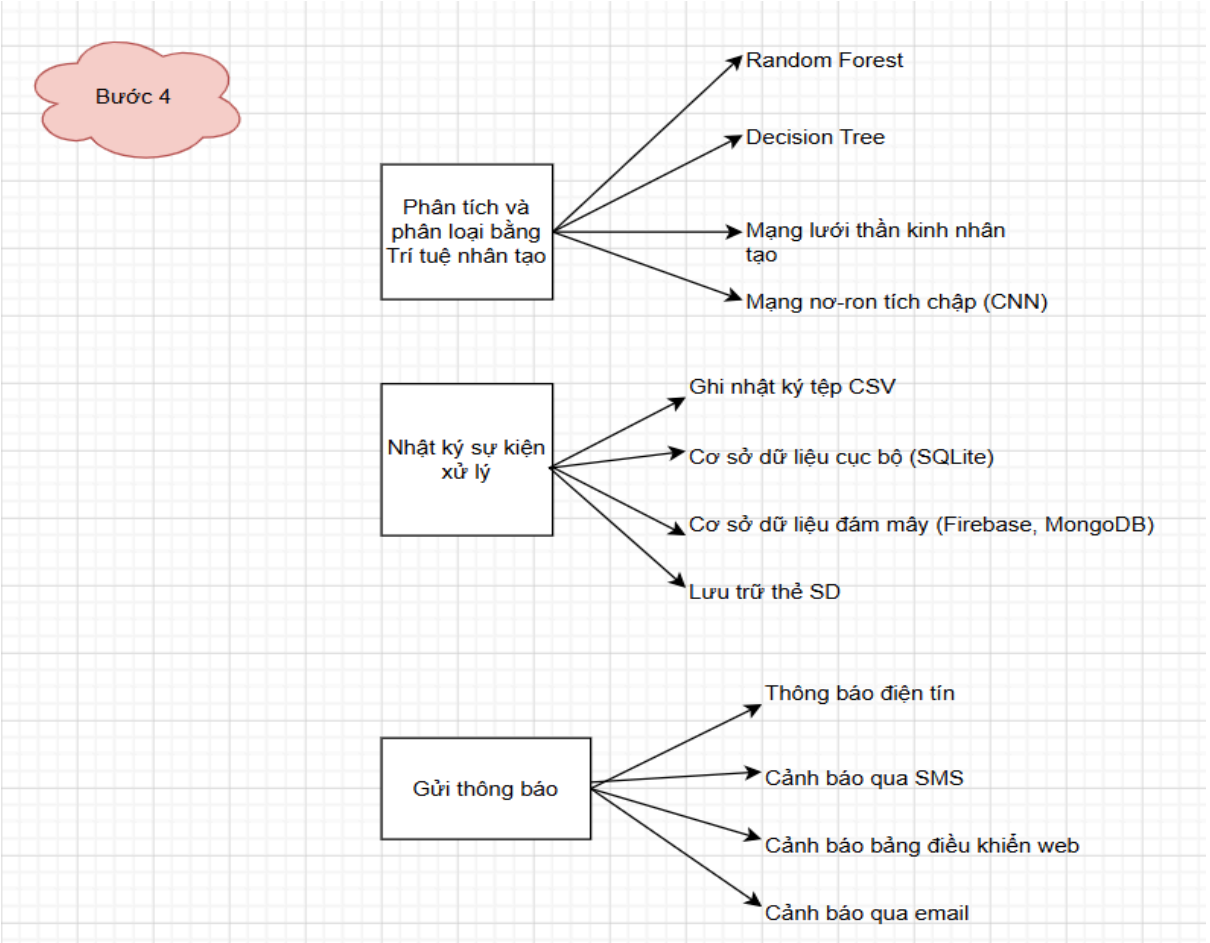


Figure 1.5: Ánh xạ các chức năng riêng lẻ tới các giải pháp vật lý khả thi

SƠ ĐỒ KHỐI (SYSTEM DIAGRAM)

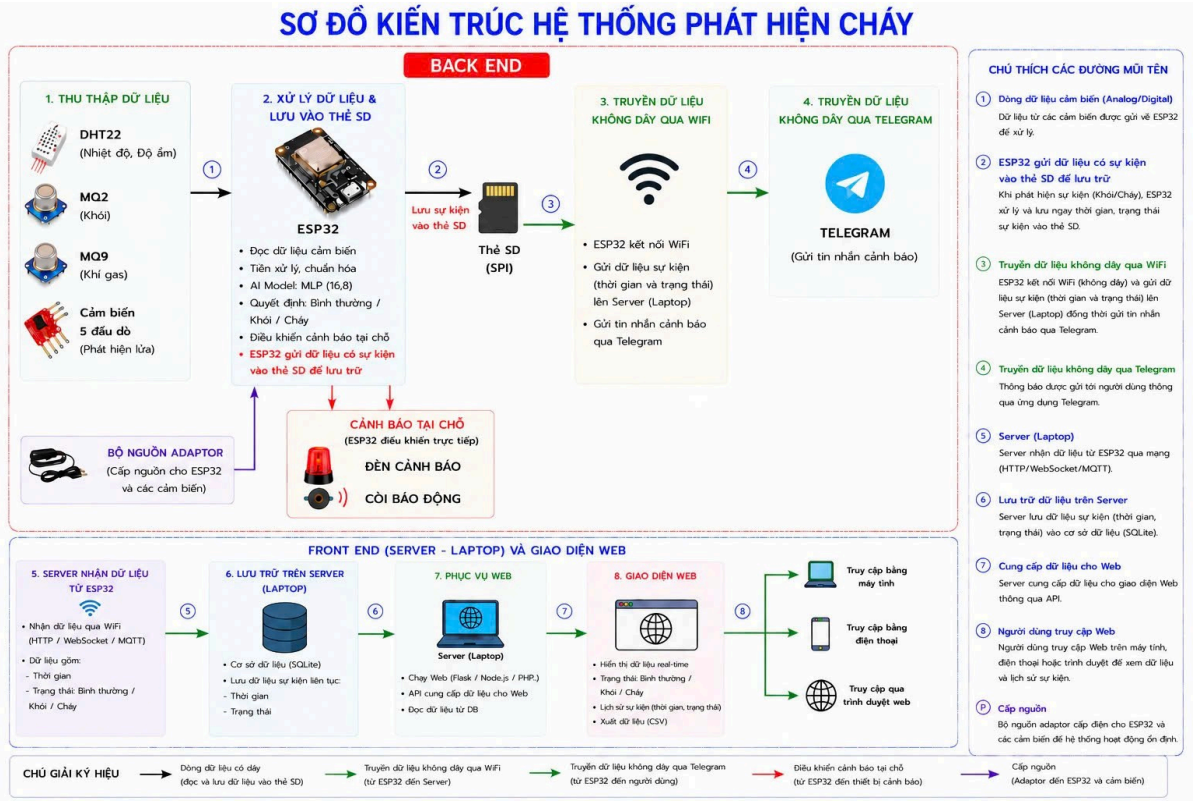


Figure 1.6: Sơ đồ hệ thống

PHÂN BIỆT FUNCTION VÀ COMPONENT

PHÂN BIỆT FUNCTIONS VÀ COMPONENTS

FUNCTIONS (HỆ THỐNG PHẢI LÀM GÌ?)	COMPONENTS (DÙNG CÁI GÌ ĐỂ LÀM?)	LOẠI
1. Thu thập dữ liệu môi trường Đo nhiệt độ, độ ẩm, khói, khí gas và phát hiện lửa	DHT22, MQ2, MQ9, Flame Sensor 5 kênh	Hardware
2. Xử lý và truyền dữ liệu Đọc, xử lý dữ liệu từ cảm biến và gửi đến hệ thống	ESP32	Hardware + Software
3. Dự đoán nguy cơ cháy Phân loại trạng thái: Normal / Smoke / Fire	MLP Model (AI)	Software (AI)
4. Lưu trữ dữ liệu Lưu dữ liệu huấn luyện và lịch sử sự kiện	SD Card, CSV File	Hardware + Software
5. Truyền dữ liệu Gửi dữ liệu đến giao diện giám sát	WiFi (ESP32)	Hardware
6. Hiển thị và giám sát Hiển thị trạng thái, biểu đồ, kết quả dự đoán	Flask Web Interface, HTML, CSS, JS	Software
7. Cảnh báo người dùng Cảnh báo âm thanh và ánh sáng khi phát hiện nguy cơ cháy	Buzzer, LED	Hardware

Figure 1.7: PHÂN BIỆT FUNCTION VÀ COMPONENT

B. Content

Stage 1 — Empathize

1. **Purpose**

- Understand who the system is for and how it will be used in real environments
- Identify user behaviors, pain points, and limitations
- Capture environmental and operational constraints
- Ensure the system is designed based on real needs, not assumptions

2. **Stakeholders**

- Primary Users: Security
  Responsibilities:
  + Monitor the system
  + Receive alerts
  + Make decisions during emergencies
- Secondary Users: Building manager, maintenance and technical staff
- Responsibilities:
  + Monitor system performance
  + Maintain and troubleshoot hardware
- Affected Users:
  + Parking users
  + Visitors
- Impact: Affected by system accuracy (false alarms or missed detection)
- Regulatory Stakeholders: Fire safety authorities
- Concerns: System reliability and safety compliance

3. **User Personas**

Personas	Security Guard Role: Managing the underground parking lot (especially during night shifts)	Maintenance Staff Role: Maintain and inspect system components	Parking Users Do not interact directly with the system
Target	Detect incidents quickly Respond immediately		
Weaknesses	Cannot continuously focus on all cameras Easily miss early fire signals Too many false alarms reduce trust	Sensors affected by dust and humidity Difficult to identify faulty sensors	
Needs	Clear and immediate alerts Minimal false alarms Accurate location of the incident	Easy system diagnostics Access to event logs Clear system status	Personal safety Avoid unnecessary panic due to false alarms
Environment	Low light Multiple screens Long working hours		

4. **Context of Use**

- Operational:
  + System operates 24/7
  + Limited staff during night shifts
  + One operator may monitor multiple zones
- Task Context:

  Typical workflow:

  System continuously monitors environment

  Abnormal signals are detected

  Alert is triggered

  Security staff receives alert

  Staff verifies situation

  Response action is taken
- Environmental Context:
  + Low-light or dark conditions
  + Dust and humidity
  + Vehicle exhaust gases (CO, CO_2)
  + Noise and distractions

5. **User problems**

- Manual monitoring may detect fires too late.
- Lack of early warning increases damage risk.
- No centralized alert and logging system exists.
- Difficult to identify the exact risk zone quickly without automated detection.
- Privacy concerns:
  + Camera monitoring may capture people's faces
  + Video data storage must follow privacy policies
  + Prefer edge processing without long-term video storage

6. **User need (Based on stakeholder analysis)**

- Early fire detection to reduce damage and risk.
- Clear and fast alerts so security staff can respond immediately.
- Reliable monitoring system that operates continuously (24/7).
- Simple interface for easy monitoring and response.
- Automatic event logging for later investigation and reporting

7. **Maintenance constraints**

- Sensors may accumulate dust
- Cameras may be blocked or dirty
- Periodic calibration required
- Maintenance staff must easily access system diagnostics

8. **Real Operating Constraints**

- Vehicle exhaust and dust may cause false positives.
- Sensor accuracy depends on placement and maintenance.
- Network failures may delay remote alerts.
- Power failures may disable parts of the system.
- Real fire testing is limited — safe simulation must be used.

Stage 2 — Define

1. **Problem Statement:**

Design a small-scale automated fire detection and alert system to simulate fire risk conditions in parking basements.

- System Objectives:
  + Detect early fire indicators such as smoke, gas, flame signals, and simplified temperature changes.
  + Provide clear and fast alerts to security staff and building management.
  + Operate continuously (24/7) for real-time monitoring.
  + Reduce false alarms caused by environmental factors such as dust, humidity, or vehicle exhaust.
  + Allow security personnel to quickly identify the location of potential fire risk.

2. **Success Metrics:**

These metrics are evaluated under controlled and simulated conditions due to the small-scale prototype implementation.

a. **Detection Speed:**

- The system should trigger an alert within 20 seconds after abnormal smoke, gas, temperature, or flame signals are detected.

b. **Detection Accuracy:**

- The system should correctly detect at least 75-80% of simulated fire events during testing.

a. **False Alarm Rate:**

- False alarms caused by dust or vehicle exhaust should remain below 5% of total alerts.

b. **Alert Response:**

- The alarm (buzzer/light/interface notification) should activate immediately after detection.
- Security staff should be able to recognize the alert within 25 seconds.

c. **System Reliability:**

- The monitoring system should operate continuously with at least 75-80% uptime during testing.

d. **Event Logging:**

- All alert events must be recorded with timestamp and system information for later monitoring and investigation.

3. **System Scope:**

- Functional Requirements (What the system does)
  + Monitor environmental conditions in parking basements using sensors
  + Detect early signs of fire risk (smoke, gas, temperature changes, flame signals)
  + Process sensor data to identify abnormal conditions
  + Generate automatic alerts (buzzer, light, or interface notification)
  + Provide information about the location of detected risk
  + Record system events and alerts in a database for monitoring and analysis

- Non-Functional Requirements (How the system performs)
  + Detection latency ≤ 20 seconds after abnormal signals are detected
  + Detection accuracy ≥ 75–80% under simulated conditions
  + False alarm rate ≤ 5% under environmental disturbances (dust, exhaust)
  + System uptime ≥ 75–80% during testing
  + The system must operate continuously (24/7 monitoring capability)
  + The system must function in low-light environments
  + The system must be robust to dust, humidity, and vehicle exhaust gases
  + Data stored in the database must be consistent, time-synchronized, and reliable
- System Boundaries (What the system does NOT do)
  + The system does not automatically extinguish fires
  + The system does not replace official fire protection systems (e.g., sprinkler systems)
  + The system does not directly contact the city fire department
  + The system operates as a supporting early-warning tool, not a full fire safety solution

4. System Specifications:

The proposed system is implemented as a small-scale prototype to simulate fire detection in parking basement environments. This section describes the system components, operating conditions, and data handling requirements.

a. Sensors:
- Smoke sensor (detect smoke-like particles)
- Gas sensor (detect gas concentration such as CO or similar gases)
- Flame sensor (detect presence of flame signals)
- Temperature sensor (detect simplified temperature changes)

b. Operating Conditions: The system must operate under conditions that simulate real parking basement environments:
- Low lighting or dark environment.
- Dust and smoke-like particles.
- Limited space and airflow.
- Continuous monitoring in real time.

c. Processing Unit:
- A microcontroller or embedded controller collects sensor data, analyzes abnormal conditions, and determines whether an alert should be triggered.

d. Alert System:
- When dangerous conditions are detected, the system activates alerts such as a buzzer, warning lights, or interface notifications to inform security staff immediately.

e. Monitoring Interface:
- The interface allows security staff to monitor system status and identify alerts quickly and easily.

f. Event logging:
- All alert events are automatically recorded in a structured database for monitoring and analysis.

g. Database system:

- The database collects data from sensors and the processing unit, including:
  + Sensor ID
  + Sensor type.
  + Timestamp.
  + Measured data.
  + System status.
  + Alert information.
  + Data label (Normal / Smoke / Fire)
- The dataset must include:
  + Normal conditions
  + Smoke conditions (e.g., incense, cigarette)
  + Fire conditions (e.g., lighter, burning paper)
- Data must be collected under multiple environmental conditions:
  + Dark environment

h. System Capability: The system is expected to:

- Detect abnormal environmental patterns related to fire risk.
- Provide faster alerts than manual monitoring.
- Reduce false alarms caused by environmental disturbances.
- Support real-time monitoring and decision-making

i. System Constraints:

- Low-light environment within the enclosed model.
- Dust and smoke-like particles affecting sensor readings.
- Limited space in the small-scale rectangular model.
- Sensor accuracy and calibration limitations.

Stage 3 — Ideate

1. Solution Concepts

a. Threshold- Based sensor detection

In this concept, environmental sensors continuously measure indicators such as smoke-like signals, gas concentration, flame signals, and simplified temperature changes within the simulated basement model.. A microcontroller collects sensor readings and compares them with predefined safety thresholds. When a sensor value exceeds its threshold, the system immediately activates an alarm.

Principle: The system continuously reads sensor measurements and compares them with predefined safety threshold values to determine whether abnormal conditions are present.

System architecture: Sensors → Microcontroller → Threshold Detection → Alarm System.

Advantages	Disadvantages
+ Simple system design + Easy to deploy on embedded systems + Low implementation cost	+ Threshold values may be affected by environmental conditions + Vehicle exhaust or dust may trigger false alarms + Limited ability to distinguish real fires from other environmental disturbances

b. Vision-Based Fire Detection Using Camera

In this concept, cameras are installed in the parking basement to monitor the environment visually. The captured images or video streams are analyzed using computer vision or artificial intelligence algorithms to detect smoke or flames. Image processing techniques or deep learning models can identify visual characteristics of fire, such as flame color patterns or smoke movement.

System architecture: Camera → Image Processing / AI Model → Fire Detection → Alarm System.

Advantages	Disadvantages
+ Early detection of visible smoke or flames + Ability to identify the location of fire incidents + Visual confirmation of alerts	+ Higher hardware cost + Performance may be affected by lighting conditions or visual obstruction + Requires additional processing power

c. Sensor fusion detection

In this concept, the fire detection system combines data from multiple environmental sensors and uses an AI-based model to improve detection accuracy. Instead of relying on a single sensor or simple threshold rules, the system analyzes patterns from several sensor readings to determine whether a fire condition is present

For example: If the system detects an increase in smoke level together with a rapid rise in temperature, the AI model may classify the situation as a potential fire risk. Similarly, if high gas concentration and flame signals are detected simultaneously, the model can confirm a fire event and activate the alarm.

→ By learning patterns from sensor data, the AI model can better distinguish real fire events from environmental disturbances such as vehicle exhaust, dust, or temporary temperature fluctuations.

Advantages	Disadvantages
+ Higher detection accuracy + Reduced false alarms + More reliable monitoring in complex environments	+ More complex system design + Higher hardware cost due to multiple sensors + Requires sensor calibration

2. Data Strategy

Data Sources: The system collects environmental data from multiple sensors installed in the parking basement.

- + Smoke sensors
- + Temperature sensors
- + Gas sensors

Data Collection: Sensor readings are collected periodically and transmitted to the processing unit.

- + Sensor ID

+ Measured value
+ Timestamp
+ System status

Data Labeling and Testing: Because real fire testing in a parking basement is unsafe. Testing is conducted in bright lighting conditions and in dark areas, as well as in high-temperature and low-temperature conditions. The experiments also consider different smoke densities, including both thick and thin smoke.

+ Artificial smoke generation
+ Controlled heat sources
+ Safe flame simulation

Data Challenges: Several challenges may affect the reliability of sensor data

+ Vehicle exhaust gases may resemble smoke signals
+ Dust particles may trigger smoke sensors
+ Sensor drift over time may reduce accuracy
+ Environmental factors such as humidity may influence readings

3. System Architecture Design

Sensor Layer: environmental sensors continuously monitor temperature, smoke levels, gas concentration, and flame signals in the parking basement. If cameras are used, they provide visual monitoring of the environment.

+ Smoke sensor
+ Temperature sensor
+ Camera

Processing Layer: a microcontroller or embedded processing unit collects sensor data and performs detection analysis (Arduino, ESP32, STM32, Raspberry Pi).

Decision Layer: the system determines whether abnormal conditions represent a potential fire risk. If the system detects dangerous conditions, it triggers the alert mechanism.

+ AI model
+ Threshold logic
+ Sensor fusion logic

Alert Layer:

+ Buzzer
+ Warning light
+ Notification (laptop, phone, control center)

Event Logging Layer: All sensor readings and alert events are recorded with timestamps. This logging system enables monitoring, analysis, and investigation of incidents

+ SD card
+ Database
+ Cloud

The system follows a multi-input → single-output (decision) architecture, combining camera and sensor data. So, the system is divided into two layers:

+ Image Processing Layer: uses YOLOv8 (YOLOv8 is used only for image processing (fire/smoke detection) and cannot handle sensor data)
+ Fusion Layer: Based on references and suggestions from AI platforms such as GPT and Gemini, suitable models for this layer include *Random Forest* and *Logistic*

Regression, which can effectively integrate multiple data sources and output a single, reliable fire detection result.

The system uses an edge computing approach for real-time fire detection. AI processing (using YOLOv8) is performed locally, and when a fire is detected, a notification is sent to smartphones via Cloud Messaging.

4. Trade-Off Analysis and Concept Selection

Based on the user needs identified in Empathize stage, a trade-off analysis table is developed to evaluate and compare the proposed solution concepts as follows

Criteria	**Threshold-Based Detection**	**Vision-Based Detection**	**Sensor Fusion Detection**
Early fire detection	Medium (50%)	High (90%)	High (90%)
Detection accuracy	Low (10%)	Medium (50%)	High (90%)
False alarm resistance	Low (10%)	Medium (50%)	High (90%)
Reliability in basement environment	Medium (50%)	Medium (50%)	High (90%)

→ Based on the trade-off analysis, Sensor Fusion Detection is selected because it provides higher detection accuracy and reliability while remaining feasible for embedded system implementation

Stage 4 — Prototype

I. System

System Architecture

The proposed system consists of:

- Smoke Sensor (MQ-2)
- Temperature and Humidity Sensor (DHT22)
- 5-Channel Flame Sensor Module
- Gas Sensor (MQ-9)
- ESP32 Embedded Controller
- Power adapter for ESP32
- Alarm and Notification System (Buzzer and LED)

Sensor Data Acquisition

The ESP32 continuously collects environmental information from multiple sensors:

- Smoke concentration
- Temperature
- Humidity
- Presence of flame
- Carbon monoxide concentration

Data Preprocessing: The collected sensor data are filtered and normalized to eliminate noise and improve system stability.

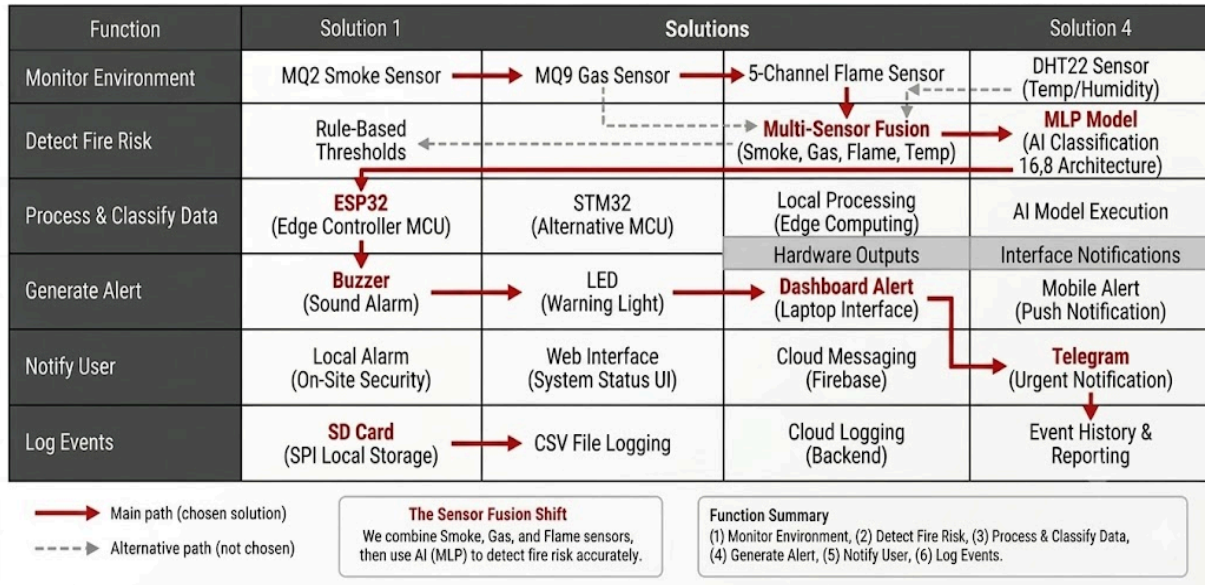


Figure: The Morphological Chart to visualizes the entire concept generation space

II. Database

1. Step 1: Dataset Identification

The database used in this project is designed for a supervised learning problem. The main objective of the dataset is to support the AI model in learning the relationship between sensor readings and fire-risk conditions. In this system, the input data are collected from hardware sensors, and the output is a class label representing the current environmental condition.

1.1 Problem Type:

This project is formulated as a multi-class classification problem. The AI model classifies each sensor data record into one of three predefined classes:

- 0 = Normal
- 1 = Smoke
- 2 = Fire

Classification is suitable for this project because the system does not predict a continuous value, but identifies the operating condition of the environment based on sensor patterns.

1.2 Data Modality:

The dataset belongs to the tabular data type. Each data sample is stored as one row in a CSV file, and each column represents a sensor feature or related information.
The main input features include:

- Timestamp
- Temperature
- Humidity
- MQ2 smoke/gas sensor value

- MQ9 gas sensor value
- Flame analog value
- Flame digital signals
- Light condition

This data modality is appropriate because the system uses numerical sensor readings rather than images, audio, or text.

1.3 Label Type:

The dataset uses discrete labels for supervised learning. Each sensor record is assigned one label according to the experimental condition during data collection.

Label	**Class**	**Meaning**
0	Normal	Safe condition, no smoke and no flame
1	Smoke	Smoke condition or early fire indicator
2	Fire	Flame or direct fire-risk condition

1.4 Data Sources:

The dataset is self-built by collecting data from the prototype hardware system. The data sources include:

Sensor / Component	**Collected Information**
DHT22	Temperature and humidity
MQ2	Smoke/gas-related analog value
MQ9	Gas/CO-related analog value
5-channel Flame Sensor	Flame analog and digital signals
ESP32	Data acquisition, timestamp handling, data transmission, logging

The dataset is therefore a custom dataset because it is collected directly from the team's own prototype instead of using a public dataset.

1.5 Suitability of the Dataset

The selected dataset is suitable for the project for the following reasons:

- It reflects the actual input signals used by the prototype system.
- It contains the three target conditions required by the fire detection task: Normal, Smoke, and Fire.
- It is collected from the same sensors used in the real system, so the training data are consistent with the deployment environment.
- It supports supervised learning because every data record has a corresponding label.

Therefore, this database is appropriate for training and evaluating an AI-based fire detection model for a small-scale parking basement prototype.

2. Step 2: Custom Dataset Construction

2.1. Data Collection Objective:

- The objective of this stage is to construct a custom dataset that accurately represents the operating conditions of an underground parking fire detection system. The collected data are intended to train the AI model to distinguish between normal conditions, smoke conditions, and fire conditions.

2.2. Data Collection Scenarios:

- Three data collection scenarios were designed:
- Normal Condition
  + Data is collected under normal operating conditions without smoke or flame. This scenario allows the model to learn the baseline behavior of the environment and helps reduce false alarms.
- Smoke Condition
  + Smoke is generated in a controlled and safe manner to simulate the early stage of a fire. This scenario enables the model to recognize abnormal conditions before flames appear.
- Fire Condition
  + A controlled flame source is used to simulate actual fire situations. This scenario provides representative samples for dangerous conditions requiring immediate warning.

2.3 Environmental Conditions:

Since the proposed system is intended for underground parking environments, the team selected a location with weak illumination to conduct data collection.
All three scenarios were collected in the same environment to ensure consistency and to better represent real deployment conditions. The low-light environment resembles the characteristics of underground parking areas where natural light is limited.

2.4 Data Acquisition Devices:

Data were collected directly from the prototype hardware described in **Step 1.4**

2.5 Data Acquisition Process

During the experiments, ESP32 continuously reads the sensor values and stores the measurements in CSV files. Each data sample contains:

- Timestamp
- Temperature
- Humidity
- MQ2 value
- MQ9 value
- Flame digital values
- Corresponding label

The acquired data are then organized into three groups:

- Normal Data
- Smoke Data
- Fire Data

2.6 Quality Control

To ensure dataset quality, the following principles were applied:

- Data were collected according to predefined scenarios.
- Sensor readings were monitored to avoid invalid measurements.
- Samples with missing values were discarded.
- Data collection procedures were performed consistently.
- All experiments were conducted under controlled and safe conditions.

The resulting output of this step is a raw custom dataset that is ready for preprocessing and labeling in the subsequent stages.

3. Step 3: Data Preprocessing and Data Cleaning:

3.1. Objective of Data Preprocessing

After completing the data collection process, the obtained dataset contains raw sensor readings collected from the prototype. Before training the AI model, preprocessing and cleaning are necessary to improve data quality and ensure that the dataset is suitable for machine learning algorithms.

The objectives of this step are:

- Remove invalid and inconsistent samples.
- Ensure data consistency.
- Improve data quality.
- Prepare the dataset for model training.

3.2. Problems Found in Raw Data

Raw sensor data may contain several issues, including:

- Missing values caused by communication interruptions.
- Unstable sensor readings due to sensor noise.
- Duplicate records generated during continuous sampling.
- Inconsistent formatting among data entries.
- Differences in numerical scales between sensor values.

These problems may negatively affect the performance of the AI model if they are not properly handled.

3.3. Data Cleaning

To improve dataset quality, several cleaning operations were performed:

- Invalid samples were removed.
- Duplicate records were discarded.
- Missing values were checked and excluded.
- Inconsistent data entries were corrected.

As a result, only valid and reliable samples were retained for model development.

3.4 Data Standardization

Since the input features have different numerical ranges, data standardization was applied before training the model.

The StandardScaler technique was employed to transform the input variables into a common scale with zero mean and unit variance. This process helps improve the convergence speed and stability of the MLP model.

3.5. Feature Selection

The following features were used as inputs to the AI model:

- Temperature
- Humidity
- MQ2 sensor value
- MQ9 sensor value
- Flame analog value
- Flame digital signals

These features provide sufficient information for distinguishing between Normal, Smoke, and Fire conditions.

3.6. Data Quality Evaluation

After preprocessing, the dataset satisfies the following requirements:

- No missing values.
- No duplicate records.
- Consistent data format.
- Standardized numerical features.
- Improved data reliability.

Therefore, the processed dataset is suitable for the subsequent stages of labeling, dataset splitting, and AI model training.

The output of this step is a cleaned and standardized dataset ready for supervised learning.

4. Step 4: Data Annotation and Labeling

4.1 Objective of Data Labeling

In supervised learning, each data sample must be associated with a corresponding label. Data labeling enables the AI model to learn the relationship between sensor measurements and environmental conditions.

The purpose of this step is to assign the correct class to each sensor record collected during the data acquisition process.

4.2 Label Schema Design

The proposed fire detection system uses three labels corresponding to three operating conditions.

Label	Class	Description
0	Normal	Normal operating condition without smoke or flame
1	Smoke	Presence of smoke indicating the early stage of a fire
2	Fire	Presence of flame representing a dangerous fire condition

The label set is mutually exclusive and covers all operating conditions considered in this project.

4.3 Annotation Guideline

To ensure consistency during the labeling process, the following rules were defined:

- Label 0 (Normal) is assigned when no smoke or flame is present.
- Label 1 (Smoke) is assigned when smoke is intentionally generated under controlled conditions without visible flame.
- Label 2 (Fire) is assigned when a flame source is introduced to simulate an actual fire event.

These rules are applied consistently throughout the entire dataset.

4.4 Annotation Process

The labeling process was performed manually based on the experimental scenarios designed during data collection.

Each row of sensor measurements stored in the CSV file was assigned a corresponding label according to the operating condition during acquisition.

Therefore, the dataset consists of labeled samples belonging to three categories:

- Normal Data
- Smoke Data
- Fire Data

4.5 Label Verification

After labeling, the dataset was reviewed to ensure:

- Label accuracy.
- Label consistency.
- No missing labels.
- Correct correspondence between sensor values and environmental conditions.

This verification process helps improve dataset quality and reduces the possibility of label noise.

4.6 Output of Step 4

The output of this step is a fully labeled dataset in which each sensor record is associated with one of the three classes:

- Normal
- Smoke
- Fire

This labeled dataset is then used for AI model training and evaluation in subsequent stages.

5. Step 5: Data Augmentation:

5.1 Objective of Data Augmentation

Data augmentation is commonly used to increase the size and diversity of datasets. The purpose of this step is to improve the generalization capability of the AI model and reduce overfitting.

However, the proposed fire detection system uses tabular sensor data instead of image data. Therefore, conventional image augmentation techniques are not applicable.

5.2 Data Augmentation Strategy

Since the dataset consists of numerical sensor readings collected from DHT22, MQ2, MQ9, and flame sensors, no artificial data augmentation techniques were applied.

Instead of generating synthetic samples, additional data were collected under different operating scenarios to improve dataset diversity and maintain class balance.

5.3 Class Balance

To avoid model bias toward a particular class, the number of samples collected for each category was maintained as balanced as possible.

The dataset includes three classes:

- Normal
- Smoke
- Fire

Balanced class distribution helps the AI model learn the characteristics of each condition more effectively and improves classification performance.

5.4 Benefits of the Adopted Approach

The adopted strategy provides several advantages:

- Preserves the authenticity of sensor measurements.
- Avoids introducing unrealistic synthetic data.
- Improves the diversity of the dataset through real-world data collection.
- Reduces the risk of overfitting.
- Enhances the generalization capability of the AI model.

5.5 Output of Step 5

The output of this step is a balanced and representative dataset with sufficient diversity for training the AI model.

No artificial data augmentation techniques were applied because the dataset is based on real sensor measurements collected from the prototype system.

6. Step 6: Dataset Splitting

6.1 Objective of Dataset Splitting

Before training the AI model, the dataset was divided into separate subsets to ensure objective model evaluation and prevent overfitting.

The dataset was split into three independent subsets:

- Training set
- Validation set
- Testing set

6.2 Splitting Ratio

The dataset was divided using the following ratio:

Dataset	Ratio	Number of samples
Training Set	70%	2709
Validation Set	10%	387
Testing Set	20%	774

The training set is used to train the AI model. The validation set is used to monitor model performance and tune parameters. The testing set is used to evaluate the final model on unseen data.

6.3 Class Distribution

To ensure fairness during training and evaluation, the class distribution was kept balanced in all subsets.

Dataset	Normal	Smoke	Fire
Training Set	903	903	903
Validation Set	129	129	129
Testing Set	258	258	258

This balanced distribution helps prevent the model from being biased toward one class.

6.4 Data Independence

Each data sample appears in only one subset. Training data are not reused in validation or testing. This prevents data leakage and ensures that the final evaluation reflects the generalization ability of the AI model.

6.5 Output of Step 6

The output of this step consists of three prepared datasets:

- fire_train_70.xlsx
- fire_validation_10.xlsx
- fire_test_20.xlsx

These datasets are used for training, validation, and final testing of the fire detection AI model.

7. Step 7: Dataset Documentation and Usage Guidelines

7.1 Dataset Overview

The dataset was developed to support the training and evaluation of an AI-based fire detection system for underground parking environments.
The dataset is designed for a supervised multi-class classification problem and consists of three classes:

- Normal
- Smoke
- Fire

The dataset contains sensor readings collected directly from the prototype system and represents realistic operating conditions.

7.2 Data Source and Collection

The dataset is self-built using the prototype hardware system.
The main data acquisition devices include:

- DHT22 sensor
- MQ2 sensor
- MQ9 sensor
- Five-channel flame sensor
- ESP32 embedded system

The data were collected under controlled experimental scenarios representing normal conditions, smoke conditions, and fire conditions.

7.3 Preprocessing Pipeline

Before training the AI model, the collected raw dataset underwent preprocessing and cleaning procedures, including:

- Removal of invalid samples.
- Elimination of duplicate records.
- Inspection of missing values.
- Standardization using StandardScaler.

These operations improve data consistency and enhance model performance.

7.4 Label Description

The dataset uses the same label schema introduced in **Step 4.2**

7.5 Data Structure

The dataset is stored in Microsoft Excel and CSV formats.
Each sample contains the following attributes:

- Timestamp
- Temperature
- Humidity
- MQ2 sensor value
- MQ9 sensor value
- Flame digital values
- Label

Each row corresponds to one sensor measurement.

7.6 Dataset Split

The dataset splitting configuration follows the procedure described in **Step 6.2**

7.7 Usage Guidelines

The dataset is intended for supervised learning and can be used to train and evaluate machine learning models for fire detection applications.
Users should:

- Maintain the original class labels.
- Apply the same preprocessing procedures before training.
- Avoid mixing training and testing data.
- Preserve the independence of the three subsets.

Following these guidelines helps ensure fair and reproducible experiments.

7.8 Dataset Limitations

Despite its effectiveness, the dataset has several limitations:

- Data were collected only in a small-scale prototype environment.
- The dataset mainly represents underground parking conditions with low illumination.
- Extreme environmental conditions were not extensively covered.
- Additional data may be required for large-scale deployment.

7.9 Ethics and Safety Considerations

All experiments were conducted under controlled and safe conditions.
The dataset does not contain personal information or sensitive data. Therefore, no privacy issues are involved in the collection process.
Safety precautions were applied during smoke and fire simulations to minimize risks and ensure experimental reliability.

7.10 Output of Step 7

The final output of this step is a documented dataset consisting of:

- fire_train_70.xlsx
- fire_validation_10.xlsx
- fire_test_20.xlsx

Together with the accompanying documentation, these datasets provide a complete foundation for training, validating, and evaluating the AI-based fire detection system.

III. AI MODEL SELECTION AND EVALUATION

Data collected from the hardware sensors (DHT22, MQ2, MQ9, 5-Channel Flame Sensor Module) was utilized to train the Artificial Intelligence models. To ensure objective assessment and technical safety, we applied the **benchmarking method** and selected **Recall** as the paramount evaluation metric.

Benchmarking Method with 3 Algorithms

- Executed three algorithms with varying levels of complexity in parallel to establish comparison baselines. This approach demonstrates the true capability and necessity of the selected core model:
  + **Zero-intelligence Baseline (Dummy Classifier):** Makes predictions based solely on the most frequent label (most_frequent). Surpassing this baseline proves that the sensor data collected from the Prototype has actual discriminative value and is not random.
  + **Statistical Baseline (Gaussian Naive Bayes):** A traditional probabilistic algorithm based on the assumption that sensors are independent. This baseline is used to prove that fire indicators (Temperature, Smoke, Gas) always share non-linear cross-correlations, thus requiring a more complex analytical structure.
  + **Core Model (Multi-Layer Perceptron - MLP 16,8):** An artificial neural network capable of automatically learning non-linear feature combinations from both analog and digital sensors. This is the primary "brain" chosen to be exported as a configuration file for hardware integration.

Benchmarking Table:

AI Model	Role in the System	Accuracy	Precision	Recall (Paramount Metric)	F2-Score
Dummy Classifier	Zero-intelligence Baseline: Predicts based solely on the most_frequent label. Surpassing this baseline proves that the sensor data collected from the prototype has actual discriminative value and is not random.	*Lowest*	*Lowest*	*Very Low* (Misses actual fire events)	*Very Low*
Gaussian Naive Bayes	Statistical Baseline: A traditional probabilistic algorithm assuming that sensors are independent. This proves that fire indicators (Temperature, Smoke, Gas) share non-linear cross-correlations, requiring a more complex analytical structure.	*Moderate*	*Moderate*	*Suboptimal* (Limited by linear assumptions)	*Moderate*
MLP (16,8)	Core Model: An artificial neural network capable of automatically learning non-linear feature combinations from all analog and digital sensors.	Highest	Optimized	Extremely High (Minimizes False Negatives)	Excellent (Approaches 1.0)

Evaluation Criteria: Optimizing Recall and F2-Score

- Because the automated fire detection system is a **safety-critical application**, we focused on optimizing the **Recall** metric for the two hazardous labels (Smoke and Fire) rather than the overall Accuracy:
  + **Preventing Type II Errors (False Negatives):** In fire prevention, a false alarm (False Positive) primarily causes inconvenience and can be manually verified. Conversely, **missing an actual fire (False Negative)** leads to catastrophic consequences for human life and property. Therefore, maximizing Recall is the ultimate goal to bring the hazard omission rate as close to zero as possible.
  + **Balancing with F2-Score:** The team additionally utilized the F2-Score (which weights Recall twice as heavily as Precision). This allows the system to maintain maximum sensitivity to fire/smoke while keeping the frequency of false alarms at an acceptable level, ensuring operational reliability in real-world scenarios.

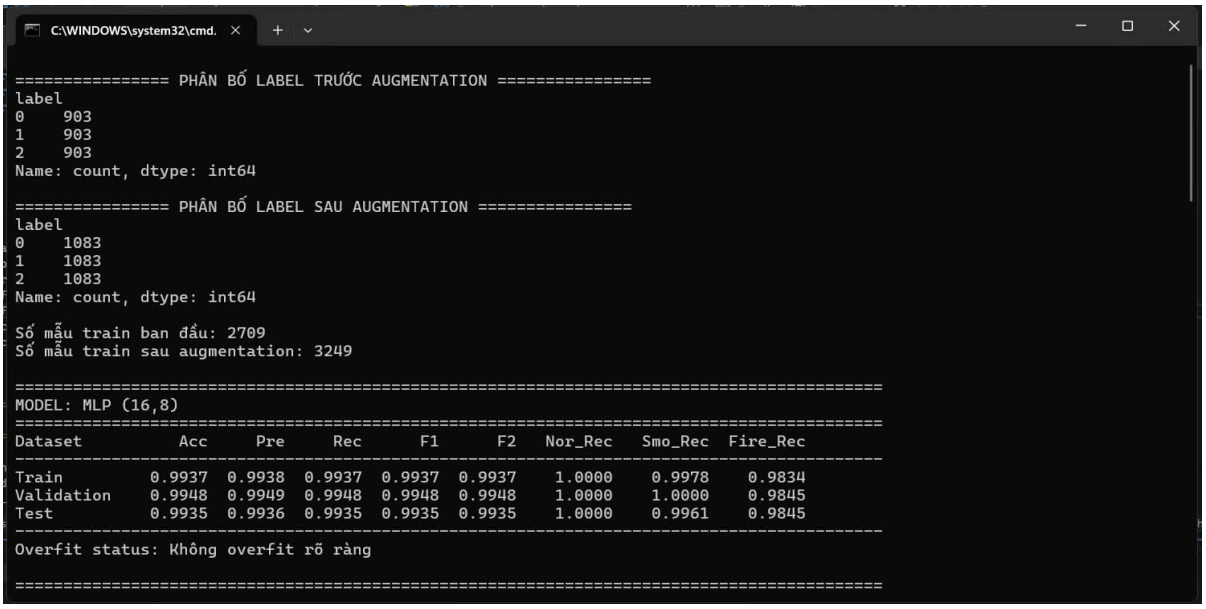


Figure 2.2

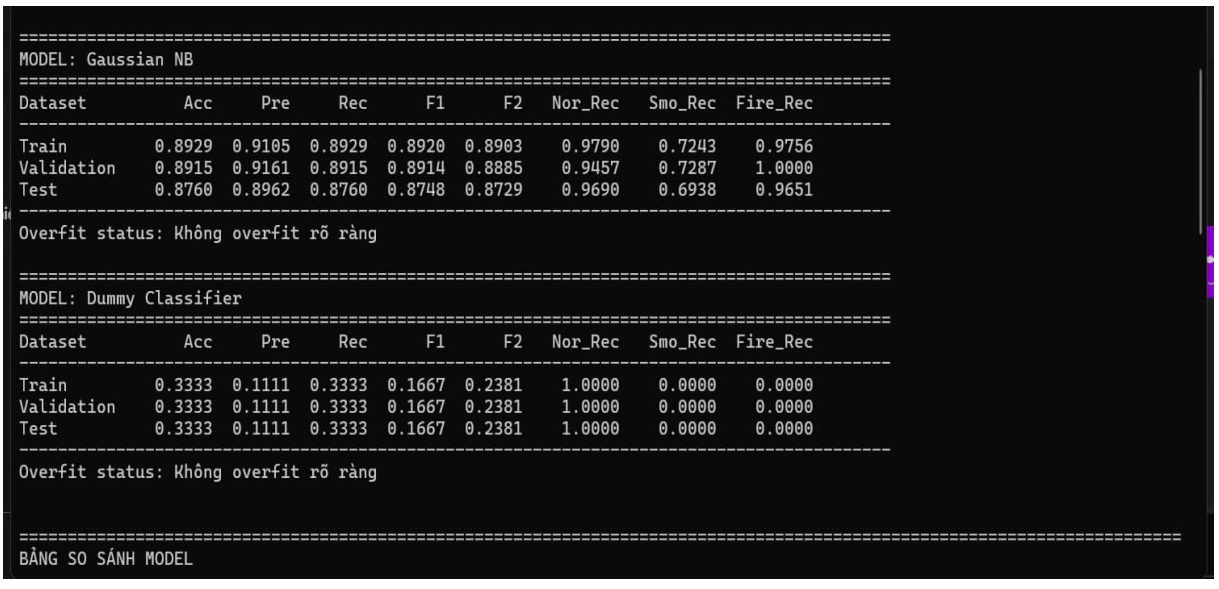


Figure 2.3

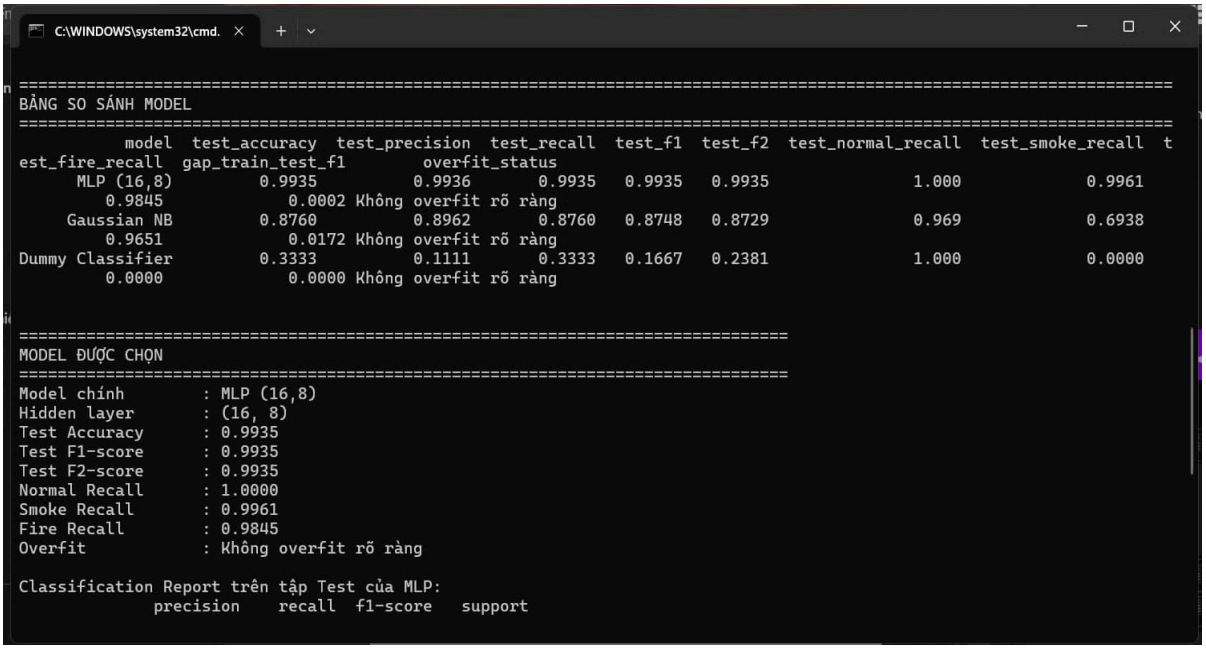


Figure 2.4

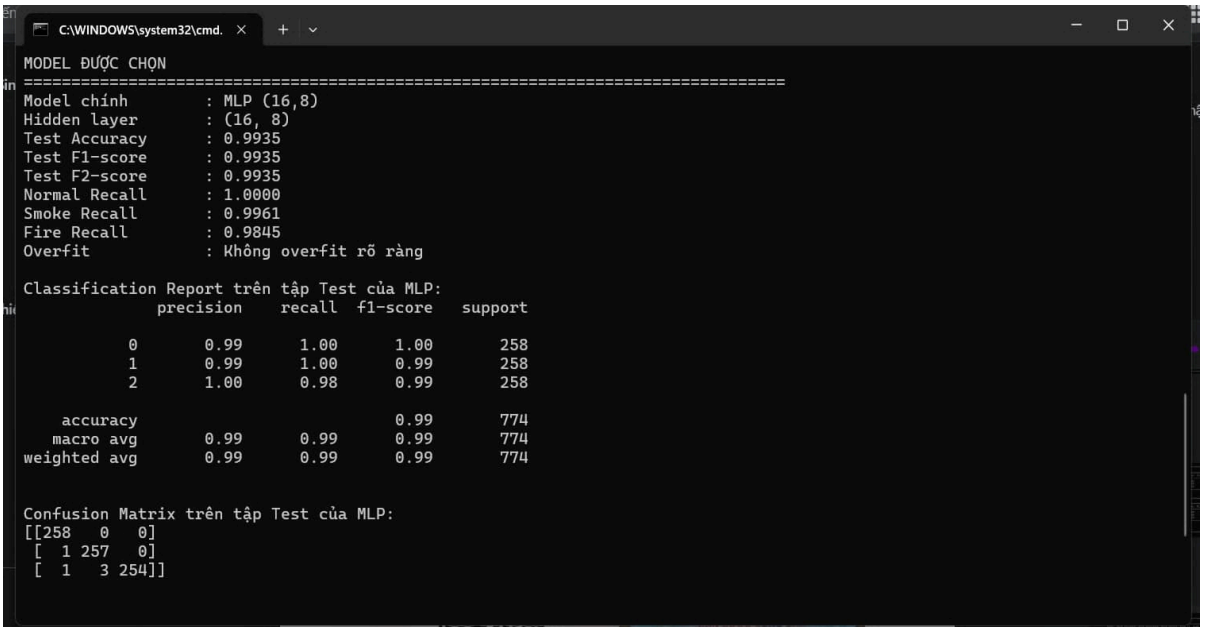


Figure 2.5

3. Justification for Selecting the MLP Model:

The **Multi-Layer Perceptron (MLP)** was selected as the core model because it effectively processes complex, non-linear feature combinations from multiple sensors. As observed in the learning curves, both the training loss and validation loss decrease smoothly and remain close to each other, indicating that the model converges properly without significant overfitting. This demonstrates an excellent generalization capability on unseen data, making it the most reliable "brain" to be exported as a configuration file for hardware integration.

Stage 5 — Test & Improve

5.1. **Core Model (MLP) Evaluation Metrics**

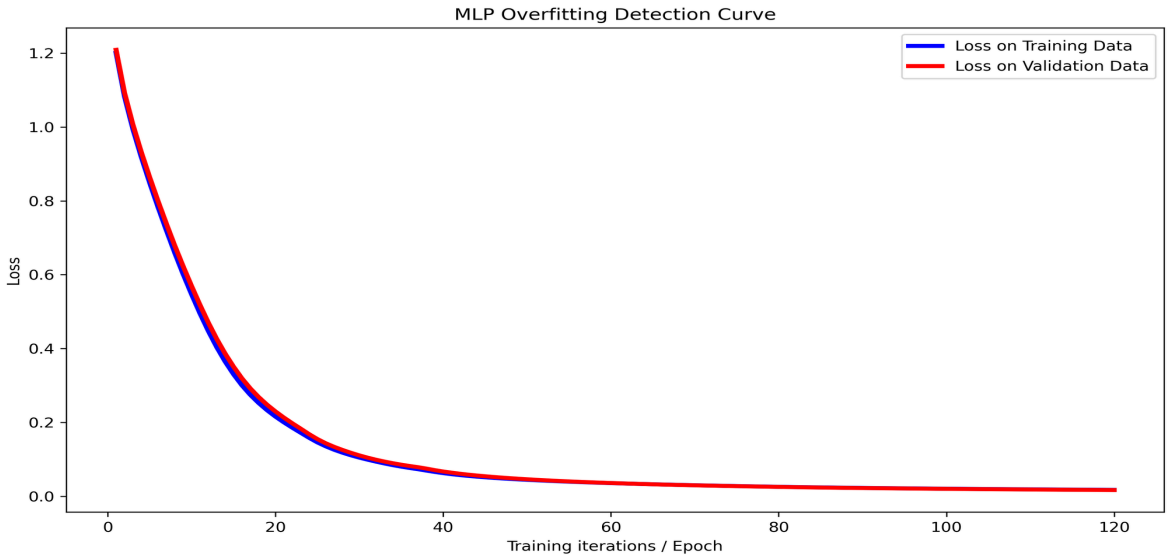


Figure 5.1: Detailed Evaluation Metrics for the Multi-Layer Perceptron (MLP) Classifier

The training loss and validation loss during the learning process of the MLP model. Both curves decrease smoothly and remain close to each other throughout training. This behavior indicates that the model converges properly and does not exhibit significant overfitting, demonstrating good generalization capability on unseen data.

5.2. **Training Dynamics and Classification Analysis**

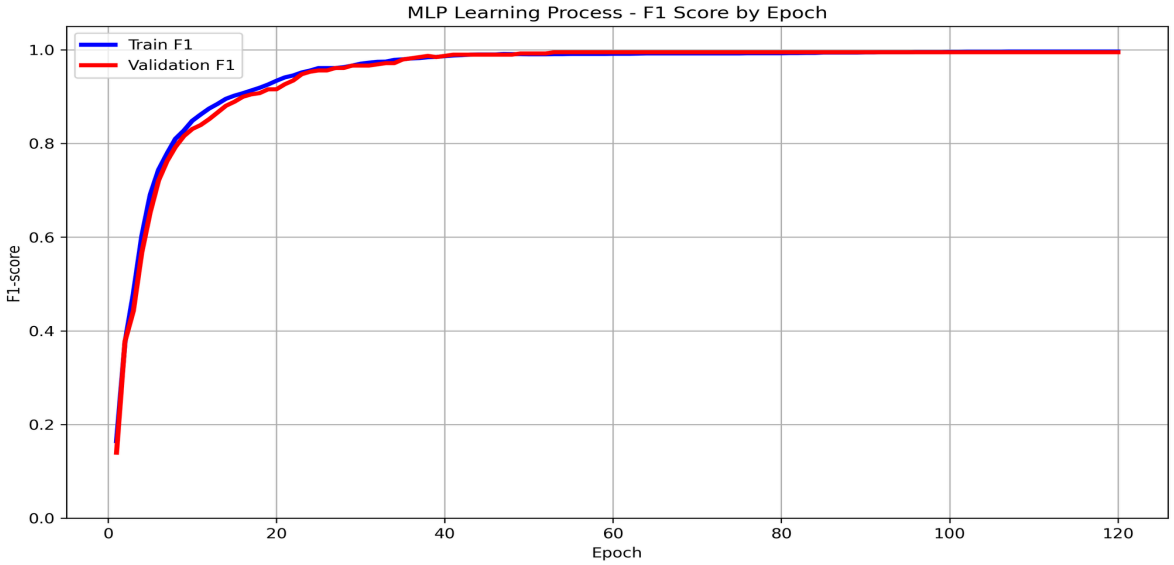


Figure 5.2: Learning Curves (Training/Validation Loss & F1-Score) and Confusion Matrix of the MLP Model

The MLP model learns effectively during the training process. The training and validation F1-score curves increase rapidly and gradually converge to values close to 1.0. Moreover, the two curves remain very close to each other throughout the training process, indicating that the model learns stably and exhibits good generalization capability without significant overfitting.

5.3. **AI Models Benchmarking and Comparison**

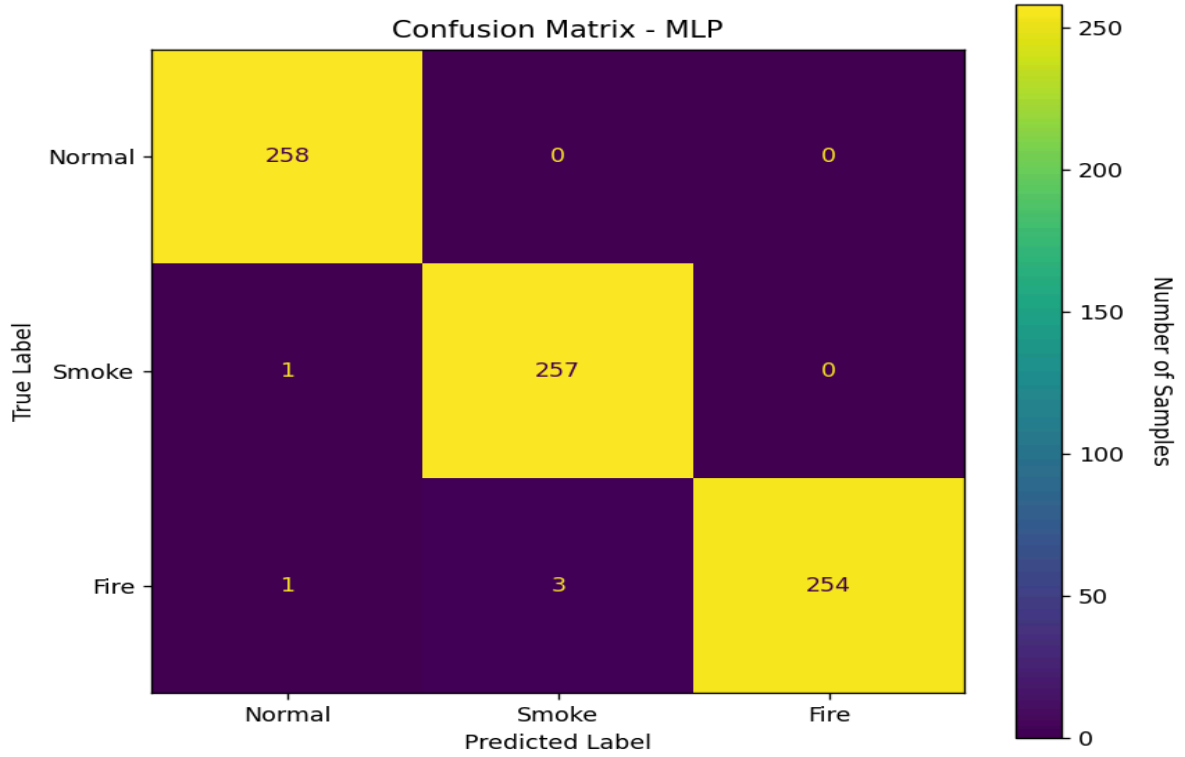


Figure 5.3: Performance Evaluation Benchmark of Baseline Models (Dummy, GaussianNB) and Core Model (MLP)

The confusion matrix of the MLP model clearly reflects the following points:

- **Outstanding correct classification rate:** The vast majority of the test samples lie on the main diagonal of the matrix, meaning they are accurately classified into the 3 independent states: Normal, Smoke, and Fire.
- **Acceptable error (Smoke → Fire):** A small number of samples belonging to the Smoke class are misclassified into the Fire class. This is entirely reasonable from a physical perspective due to the strong similarity between dense smoke concentration and early flame conditions. This misclassification does not pose a danger because the system still activates the safety alert.
- **No critical confusion:** There is absolutely no significant misclassification between the Normal class and the Fire class (i.e., there is no scenario where a severe fire occurs but the system reports normal conditions). This completely minimizes dangerous False-Negative cases.

5.4. Model's performance metrics: Accuracy, Precision, Recall, F1-score, F2-score

According to the learning curves and mixed matrix in the report, the model's performance metrics are determined and can be displayed as follows:

- Accuracy (Overall Accuracy): Achieved a very high rate, with the majority of samples belonging to all three classes (Normal, Smoke, Fire) being correctly classified into their correct labels.
- Precision: Achieved an optimal level. Specifically, the system did not confuse the Normal state with the Fire state.
- Recall (Sensitivity / Detection Rate): This is the best optimization indicator for the Fire class. The system minimized false negatives (false negatives - fire present but missed), ensuring absolute safety.
- F1 Score: The F1 curve—the score of both the trainer (Training) and the test (Verification)—grows very rapidly from the early stages and converges stably at a value close to 1.0.
- F2 Score: The system's highly effective real-world light response. In the firing scenario, Recall is more important than Precision (better to give a false alarm than to miss a fire), so the system's F2 score is extremely impressive, effectively eliminating False Negatives.

5.5. System Performance Evaluation

The system delivers reliable, real-time performance in simulated underground parking environments. The seamless data acquisition via the ESP32 combined with the swift prediction speed of the MLP model ensures that the system meets the requirement of triggering an alert rapidly after abnormal signals are detected. The localized edge processing guarantees that hardware alerts (buzzer and LED) and software notifications operate consistently with minimal latency.

5.6. ADVANTAGES AND LIMITATIONS

Advantages: The system achieves high detection accuracy and minimizes the risk of missing actual fire events (False Negatives).

- By utilizing Sensor Fusion Detection and an AI model, the system successfully reduces false alarms caused by environmental disturbances (e.g., vehicle exhaust, dust) compared to traditional threshold-based systems.
- The model exhibits excellent generalization without overfitting.

Limitations:

- The current implementation is a small-scale prototype evaluated under controlled and simulated conditions.
- There is a minor classification overlap between dense smoke and early fire states due to sensor data similarities.
- Hardware sensors may accumulate dust and require periodic maintenance and calibration to maintain long-term accuracy.

Stage 6 — Future Orientation & Conclusion

Future works:

- To transition from a small-scale prototype to an industrial-grade solution, future developments will focus on:
    + **Scaling Up:** Expanding the system architecture to monitor full-sized, multi-level underground parking basements.
    + **Algorithm & Sensor Optimization:** Integrating dedicated gas sensors (such as high-precision CO sensors) and fine-tuning the AI model to completely eliminate the confusion between smoke and early-stage fire.
    + **IoT & Automated Response Integration:** Connecting the edge-computing alerts directly to the building's automated fire suppression infrastructure (e.g., sprinkler systems, exhaust fans) and centralizing data logging on a robust Cloud dashboard for building management.

Conclusion:

- This project successfully designed and implemented a small-scale automated fire detection and alert system using a Multi-Layer Perceptron (MLP) model to simulate fire risk conditions in parking basements.
- By establishing baselines with Dummy and GaussianNB classifiers, the team proved the necessity and superiority of the MLP architecture. Prioritizing the Recall and F2-Score metrics ensured that the system is highly sensitive to hazards, virtually eliminating catastrophic false-negative cases.
- The successful integration of hardware sensors, microcontroller processing, and AI-driven classification provides a highly reliable, low-cost foundation for advancing smart fire safety systems in real-world urban infrastructure.

Video Demo: Demo

Reference:

Gaussian Naive Bayes

DummyClassifier

Multi-layer Perceptron

Detect overfit, underfit and fit

Design Thinking

How To Build Database